

Module 4 Assignment: Forensic Ballistics

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Lesson A: Firearm Basics

1. Describe two differences between a bullet and a cartridge. (2 marks)

2. Discuss why bullets and their casings are often found in different locations at crime scenes. (2 marks)

3. Using your knowledge of velocity, kinetic energy, and trajectory, explain how a handgun round becomes less effective than a rifle bullet over long distances. (3 marks)

4. Explain why a police tactical unit always takes at least one assault rifle or sniper rifle when responding to a call involving a suspect armed with a gun. (2 marks)

Lesson B: Ballistic Fingerprinting and Wound Ballistics

5. Identify three unique characteristics that a forensic ballistic expert might analyze while examining a bullet and a cartridge casing to determine the type of gun used at a crime scene. (3 marks)

6. Police officers are trained to shoot at centre mass when faced with lethal threat encounters. Centre mass is considered to be the centre of the chest or the breastbone region. Use your knowledge of wound ballistics to explain why they are trained to aim for this part of the human

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body. (2 marks)

Use the following information to answer question 7.

A 20-year old male armed with a handgun shot a rival gang member twice at close range. The victim suffered two large gunshot wounds—one to his left arm and a second wound that severed the large femoral artery in the upper portion of his left leg. Each bullet expanded upon entry and neither bullet exited the victim's body.

7. a. Identify the type of bullet that likely caused these injuries. Explain your choice. (2 marks).

b. Explain which of the two gunshot wounds most likely caused the victim's death. (2 mark)

8. Discuss how the injuries to President Kennedy and Governor Connally would have been different if Lee Harvey Oswald had used a handgun instead of a rifle. (3 marks)

Lesson C. Testing for Gunshot Residue

Use the information below to answer questions 9 and 10.

A piece of gauze turns a scarlet red colour after being swabbed over the hands of a murder suspect.

9. a. Before it was swabbed over the suspect's hands, what substance was the gauze dipped into? (1 mark)

b. What specific element was detected in this test?. (1 mark)

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- c. Identify whether the element detected in this test was from a propellant or a primer. Justify your choice. (2 marks)

10. Why is the Scanning Electron Microscope (SEM) being used more frequently in GSR testing? (2 marks)

Both the Paraffin test and Harrison-Gilroy test are performed on an armed robbery suspect. The paraffin test is positive for nitrites, and the Harrison-Gilroy test is positive for antimony. X-ray analysis with a scanning electron microscope confirms the results of both tests.

11. Can the combined results of these three tests ensure a conviction in court? (3 marks)

Lesson D. Case Study Involving Forensic Ballistics

Case Study #1: The Beltway Snipers

12. Explain why no intact bullets were found at any of the Beltway sniper shootings. (3 marks)

Military snipers are trained to shoot at and hit targets from great distances, typically 500 to 1000 metres.

13. Considering the range at which most of the shootings took place, why have some military experts criticized the media's use of the label *Beltway Sniper*? (3 marks)

Case Study #2: Distance, Shielding, and Movement Critical Incident Response

14. A security guard arrests a man at a shopping mall after he is caught trying to steal clothing. When police arrive, they suspect he is one of the three suspects from the convenience store robbery and murder. How could GSR testing help to confirm their suspicions? (3 marks)

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15. Describe three reasons police officers confronting an armed suspect should know the type of firearm the suspect is carrying (3 marks)